

2 Unit-I: Groups

Definition 2.1 (Group). A **group** is a nonempty set G together with a binary operation $*$ such that, for all $a, b, c \in G$, the following axioms hold:

1. **Closure:** $a * b \in G$.
2. **Associativity:** $(a * b) * c = a * (b * c)$.
3. **Identity element:** There exists an element $e \in G$ such that $e * a = a * e = a$ for every $a \in G$.
4. **Inverse element:** For every $a \in G$, there exists an element $a^{-1} \in G$ such that $a * a^{-1} = a^{-1} * a = e$.

The group is denoted by $(G, *)$.

If, in addition,

5. $a * b = b * a$, for all $a, b \in G$,

then $(G, *)$ is called an **abelian group** or a **commutative group**.

(Question 1 & 2-QB)

Example 2.1.

1. The set $(\mathbb{Z}, +)$ of Integers is an abelian group under usual addition ‘+’.
2. The set $(\mathbb{Q}^\times = \mathbb{Q} \setminus \{0\}, \cdot)$ of non-zero Rational numbers is an abelian group under usual multiplication ‘ $\cdot$ ’.
3. The set of Natural numbers $\mathbb{N} = \{1, 2, 3, \dots\}$ under usual addition ‘+’ is **not** a group.

(Question 1 & 2-QB)

Problem 2.1. If $\mathbb{Q}_0$ is the set of all rational numbers other than zero and $*$ is a binary operation on $\mathbb{Q}_0$ defined by $a * b = \frac{ab}{5}$, $\forall a, b \in \mathbb{Q}_0$, then prove that $(\mathbb{Q}_0, *)$ forms an abelian group.

(Question 21-QB)

Solution

1. Closure:

Let $a, b \in \mathbb{Q}_0$. Since a and b are nonzero rational numbers, $ab \neq 0$. Also, $\frac{ab}{5} \in \mathbb{Q}$ and $\frac{ab}{5} \neq 0$. Therefore, $a * b = \frac{ab}{5} \in \mathbb{Q}_0$.

Hence, $\mathbb{Q}_0$ is closed under $*$.

2. Associativity:

Let $a, b, c \in \mathbb{Q}_0$. Then $(a * b) * c = \left(\frac{ab}{5}\right) * c$.

Using the definition of $*$, we have $(a * b) * c = \frac{\left(\frac{ab}{5}\right) c}{5} = \frac{abc}{25}$.

$$\text{Similarly, } a * (b * c) = a * \left(\frac{bc}{5}\right) = \frac{a\left(\frac{bc}{5}\right)}{5} = \frac{abc}{25}.$$

Thus, $(a * b) * c = a * (b * c)$.

Therefore, $*$ is associative.

3. Identity:

Let $e \in \mathbb{Q}_0$ be the identity element. Then $a * e = a$ for every $a \in \mathbb{Q}_0$.

By the definition of $*$, we have $\frac{ae}{5} = a \implies ae = 5a$.

Since $a \neq 0$, $e = 5$.

Also, $a * 5 = \frac{5a}{5} = a$ and $5 * a = \frac{5a}{5} = a$.

Hence, the identity element is $\boxed{e = 5}$.

4. Inverse:

Let $a \in \mathbb{Q}_0$. We seek $a^{-1} \in \mathbb{Q}_0$ such that

$a * a^{-1} = 5$, where 5 is the identity element.

Thus, $\frac{a a^{-1}}{5} = 5$.

Therefore, $a a^{-1} = 25$, and hence $a^{-1} = \frac{25}{a}$.

Since $a \neq 0$, $\frac{25}{a} \in \mathbb{Q}_0$.

Moreover, $a * \frac{25}{a} = \frac{a\left(\frac{25}{a}\right)}{5} = \frac{25}{5} = 5$.

Similarly, $\frac{25}{a} * a = 5$.

Hence every element $a \in \mathbb{Q}_0$ has an inverse given by $\boxed{a^{-1} = \frac{25}{a}}$.

5. Commutativity:

For $a, b \in \mathbb{Q}_0$, it is given that $a * b = \frac{ab}{5}$.

Since ordinary multiplication of rational numbers is commutative, $ab = ba$.

Therefore, $a * b = \frac{ab}{5} = \frac{ba}{5} = b * a$.

Hence $*$ is commutative.

Therefore,

$$\boxed{(\mathbb{Q}_0, *) \text{ is an abelian group.}}$$

Problem 2.2. Show that the set $\mathbb{Q}_{-1}$ of all rational numbers except -1 , with operation defined by $a * b = a + b + ab$, $\forall a, b \in \mathbb{Q}_{-1}$, is an abelian group.

(Question 22-QB)

Solution

1. Closure:

Let $a, b \in \mathbb{Q}_{-1}$. Since $a, b \in \mathbb{Q}$, $a + b + ab \in \mathbb{Q}$.

It remains to show that $a + b + ab \neq -1$.

Suppose, for contradiction, that $a + b + ab = -1$.

Then $ab + a + b + 1 = 0$.

Factoring, we get $(a + 1)(b + 1) = 0$.

Therefore, $a + 1 = 0$ or $b + 1 = 0$, which gives $a = -1$ or $b = -1$.

But $a, b \in \mathbb{Q}_{-1}$, so neither a nor b is equal to -1 . This is a contradiction.

Hence, $a * b = a + b + ab \neq -1$.

Therefore, $a * b \in \mathbb{Q}_{-1}$.

Hence, $\mathbb{Q}_{-1}$ is closed under $*$.

2. Associativity:

Let $a, b, c \in \mathbb{Q}_{-1}$. Then $(a * b) * c = (a + b + ab) * c$.

Using the definition of $*$, we have $(a * b) * c = (a + b + ab) + c + (a + b + ab)c$.

Therefore, $(a * b) * c = a + b + c + ab + ac + bc + abc$.

Similarly, $a * (b * c) = a * (b + c + bc)$.

Using the definition of $*$, we have $a * (b * c) = a + (b + c + bc) + a(b + c + bc)$.

Therefore, $a * (b * c) = a + b + c + bc + ab + ac + abc = a + b + c + ab + ac + bc + abc$.

Thus, $(a * b) * c = a * (b * c)$.

Therefore, $*$ is associative.

3. Identity:

Let $e \in \mathbb{Q}_{-1}$ be the identity element. Then $a * e = a$ for every $a \in \mathbb{Q}_{-1}$.

By the definition of $*$, we have $a + e + ae = a$.

Therefore, $e + ae = 0$, or $e(1 + a) = 0$.

Since $a \neq -1$, $1 + a \neq 0$.

Hence, $e = 0$.

Also, $a * 0 = a + 0 + a(0) = a$ and $0 * a = 0 + a + 0(a) = a$.

Hence, the identity element is $\boxed{e = 0}$.

4. Inverse:

Let $a \in \mathbb{Q}_{-1}$. We seek $a^{-1} \in \mathbb{Q}_{-1}$ such that

$a * a^{-1} = 0$, where 0 is the identity element.

Thus, $a + a^{-1} + aa^{-1} = 0 \implies a^{-1}(1 + a) = -a$.

Since $a \neq -1$, $1 + a \neq 0$.

Hence, $a^{-1} = -\frac{a}{1 + a}$.

Since $a \in \mathbb{Q}$ and $a \neq -1$, $-\frac{a}{1 + a} \in \mathbb{Q}$.

We must also show that $-\frac{a}{1 + a} \neq -1$.

Suppose $-\frac{a}{1 + a} = -1$.

Then $-a = -(1 + a)$, which implies $-a = -1 - a$, and hence $0 = -1$, which is impossible.

Therefore, $-\frac{a}{1 + a} \neq -1$.

Thus, $-\frac{a}{1 + a} \in \mathbb{Q}_{-1}$.

Moreover, $a * \left(-\frac{a}{1 + a}\right) = a - \frac{a}{1 + a} - a\frac{a}{1 + a}$.

Therefore, $a * \left(-\frac{a}{1 + a}\right) = \frac{a(1 + a) - a - a^2}{1 + a} = 0$.

Similarly, $\left(-\frac{a}{1 + a}\right) * a = 0$.

Hence every element $a \in \mathbb{Q}_{-1}$ has an inverse given by $a^{-1} = -\frac{a}{1+a}$.

5. Commutativity:

For $a, b \in \mathbb{Q}_{-1}$, it is given that $a * b = a + b + ab$.

Since ordinary addition and multiplication of rational numbers are commutative, we have $a + b = b + a$ and $ab = ba$.

Therefore, $a * b = a + b + ab = b + a + ba = b * a$.

Hence $*$ is commutative.

Therefore,

$$(\mathbb{Q}_{-1}, *) \text{ is an abelian group.}$$

General Properties of Groups

Note. For simplicity, throughout these notes we often write ab instead of $a * b$. Here, ab denotes the group operation $*$ and does not necessarily represent ordinary multiplication.

Problem 2.3. Prove that identity element in a group is unique.

(Question 4-QB)

Solution

Suppose that e and e' are two identity elements of G . Then

$$\begin{aligned} e &= ee' & (\because e' \text{ is an identity element}), \\ ee' &= e' & (\because e \text{ is an identity element}). \end{aligned}$$

Hence $e = e'$. Therefore the identity element of a group is unique.

Problem 2.4. Prove that inverse of every element in a group is unique.

(Question 5-QB)

Solution

Let $a \in G$, and suppose that both b and c are inverses of a . Then by the inverse axiom

$$ab = ba = e \quad \text{and} \quad ac = ca = e.$$

Now,

$$\begin{aligned}
 b &= be & (\because \text{identity axiom}) \\
 &= b(ac) & (\because ac = e, \text{ inverse axiom}) \\
 &= (ba)c & (\because \text{associativity}) \\
 &= ec & (\because ba = e, \text{ inverse axiom}) \\
 &= c & (\because \text{identity axiom}).
 \end{aligned}$$

Thus $b = c$. Hence the inverse of a is unique.

Problem 2.5. Prove that in a group $(G, *)$, $(a^{-1})^{-1} = a$, for every $a \in G$.

(Question 6-QB)

Solution

By the inverse axiom,

$$a^{-1}a = e.$$

Therefore,

$$\begin{aligned}
 (a^{-1})^{-1}(a^{-1}a) &= (a^{-1})^{-1}e & (\because \text{multiplying both sides on the left by } (a^{-1})^{-1}) \\
 [(a^{-1})^{-1}a^{-1}]a &= (a^{-1})^{-1} & (\because \text{associativity and identity axiom}) \\
 (e)a &= (a^{-1})^{-1} & (\because \text{since } (a^{-1})^{-1} \text{ is the inverse of } a^{-1}) \\
 a &= (a^{-1})^{-1} & (\because \text{identity axiom}).
 \end{aligned}$$

Hence,

$$\boxed{(a^{-1})^{-1} = a}.$$

Problem 2.6. Let G be a group. Then for all $a, b \in G$, prove that $(ab)^{-1} = b^{-1}a^{-1}$. That is, prove that the inverse of a product is the product of the inverses taken in the reverse order.

(Question 28-QB)

Solution

Let $a, b \in G$. We show that $b^{-1}a^{-1}$ is both a right and a left inverse of ab .

First,

$$\begin{aligned}
 (ab)(b^{-1}a^{-1}) &= a(bb^{-1})a^{-1} & (\because \text{associativity}) \\
 &= aea^{-1} & (\because bb^{-1} = e, \text{ inverse axiom}) \\
 &= aa^{-1} & (\because \text{identity axiom}) \\
 &= e & (\because \text{inverse axiom}).
 \end{aligned}$$

Also,

$$\begin{aligned}
 (b^{-1}a^{-1})(ab) &= b^{-1}(a^{-1}a)b && (\because \text{associativity}) \\
 &= b^{-1}eb && (\because a^{-1}a = e, \text{ inverse axiom}) \\
 &= b^{-1}b && (\because \text{identity axiom}) \\
 &= e && (\because \text{inverse axiom}).
 \end{aligned}$$

Hence $b^{-1}a^{-1}$ is the inverse of ab . By uniqueness of inverse,

$$\boxed{(ab)^{-1} = b^{-1}a^{-1}}.$$

Problem 2.7. Let G be a group and $a, b, c \in G$. Then prove that

$$ab = ac \implies b = c \quad (\text{left cancellation law}),$$

and

$$ba = ca \implies b = c \quad (\text{right cancellation law}).$$

(Question 8-QB)

Solution

Suppose first that $ab = ac$. Multiplying both sides on the left by a^{-1} , we get

$$\begin{aligned}
 a^{-1}(ab) &= a^{-1}(ac) && (\because \text{equality is preserved under left multiplication}) \\
 (a^{-1}a)b &= (a^{-1}a)c && (\because \text{associativity}) \\
 eb &= ec && (\because a^{-1}a = e, \text{ inverse axiom}) \\
 b &= c && (\because \text{identity axiom}).
 \end{aligned}$$

Thus the left cancellation law holds.

Now suppose that $ba = ca$. Multiplying both sides on the right by a^{-1} , we obtain

$$\begin{aligned}
 (ba)a^{-1} &= (ca)a^{-1} && (\because \text{equality is preserved under right multiplication}) \\
 b(aa^{-1}) &= c(aa^{-1}) && (\because \text{associativity}) \\
 be &= ce && (\because aa^{-1} = e, \text{ inverse axiom}) \\
 b &= c && (\because \text{identity axiom}).
 \end{aligned}$$

Hence the right cancellation law also holds.

Problem 2.8. If $a, b \in G$, then prove that each of the equations

$$ax = b \quad \text{and} \quad ya = b$$

has a unique solution in G .

(Question 29-QB)

Solution

Consider first the equation $ax = b$. Since $a^{-1}, b \in G$, closure gives $a^{-1}b \in G$. Put

$$x = a^{-1}b.$$

Then

$$\begin{aligned} ax &= a(a^{-1}b) \\ &= (aa^{-1})b && (\because \text{associativity}) \\ &= eb && (\because aa^{-1} = e, \text{ inverse axiom}) \\ &= b && (\because \text{identity axiom}). \end{aligned}$$

Thus $x = a^{-1}b$ is a solution.

To prove uniqueness, suppose x_1 and x_2 are two solutions. Then

$$ax_1 = b = ax_2,$$

so $ax_1 = ax_2$. By the left cancellation law,

$$x_1 = x_2.$$

Hence the solution is unique, and

$$\boxed{x = a^{-1}b}.$$

Now consider $ya = b$. Since $b, a^{-1} \in G$, closure gives $ba^{-1} \in G$. Put

$$y = ba^{-1}.$$

Then

$$\begin{aligned} ya &= (ba^{-1})a \\ &= b(a^{-1}a) && (\because \text{associativity}) \\ &= be && (\because a^{-1}a = e, \text{ inverse axiom}) \\ &= b && (\because \text{identity axiom}). \end{aligned}$$

Thus $y = ba^{-1}$ is a solution.

If y_1 and y_2 are two solutions, then

$$y_1a = b = y_2a,$$

so $y_1a = y_2a$. By the right cancellation law,

$$y_1 = y_2.$$

Therefore the solution is unique, and

$$\boxed{y = ba^{-1}}.$$

Other Problems

Problem 2.9. Prove that if $a^2 = e$ for every $a \in G$, then G is abelian. (OR)

Prove that in a group G , if every element has its own inverse, then G is abelian. (OR)

Prove that in a group G , if for every $a \in G$, $a^{-1} = a$, then G is abelian
(Question 10-QB)

Solution

For every $a \in G$,

$$a^2 = e \implies aa = e.$$

Hence a is its own inverse, and by uniqueness of inverse,

$$a^{-1} = a.$$

Similarly, $b^{-1} = b$ for every $b \in G$.

Now let $a, b \in G$. Since $ab \in G$ by closure, the hypothesis gives

$$(ab)^2 = e.$$

Hence ab is its own inverse:

$$(ab)^{-1} = ab.$$

But,

$$\begin{aligned} (ab)^{-1} &= b^{-1}a^{-1} && (\because \text{inverse of a product}) \\ &= ba && (\because a^{-1} = a \text{ and } b^{-1} = b). \end{aligned}$$

Therefore $ab = ba$ for all $a, b \in G$. Hence G is abelian.

Problem 2.10. Show that, for a group G , $(ab)^2 = a^2b^2$ for all $a, b \in G$ if and only if G is abelian.

(Question 7-QB)

Solution

Suppose first that G is abelian. Then $ba = ab$, and hence

$$\begin{aligned} (ab)^2 &= abab && (\because \text{definition of square}) \\ &= aabb && (\because ba = ab, \text{commutativity}) \\ &= a^2b^2 && (\because \text{definition of powers}). \end{aligned}$$

Thus $(ab)^2 = a^2b^2$.

Conversely, suppose that

$$(ab)^2 = a^2b^2$$

for all $a, b \in G$. Then

$$\begin{aligned} abab &= aabb & (\because \text{expanding both sides}) \\ bab &= abb & (\because \text{left cancellation by } a) \\ ba &= ab & (\because \text{right cancellation by } b). \end{aligned}$$

Therefore $ab = ba$ for all $a, b \in G$, so G is abelian.

Problem 2.11. Prove that a group G is abelian if $b^{-1}a^{-1}ba = e$ for all $a, b \in G$.

Solution

Let $a, b \in G$. By hypothesis,

$$b^{-1}a^{-1}ba = e.$$

Multiplying both sides on the left by b , we get

$$\begin{aligned} b(b^{-1}a^{-1}ba) &= be \\ (bb^{-1})a^{-1}ba &= b & (\because \text{associativity and identity axiom}) \\ a^{-1}ba &= b & (\because bb^{-1} = e, \text{ inverse axiom}). \end{aligned}$$

Now multiply both sides on the left by a :

$$\begin{aligned} a(a^{-1}ba) &= ab \\ (aa^{-1})ba &= ab & (\because \text{associativity}) \\ ba &= ab & (\because aa^{-1} = e \text{ and identity axiom}). \end{aligned}$$

Thus $ab = ba$ for all $a, b \in G$. Hence G is abelian.

Problem 2.12. If G is a group of even order, prove that there exists an element $a \neq e$ such that $a^2 = e$.

Solution

Let $|G| = 2n$ for some positive integer n . We argue by contradiction. Assume that no element other than e satisfies $a = a^{-1}$.

The identity satisfies $e = e^{-1}$. For every non-identity element $a \in G$, our assumption gives

$$a \neq a^{-1}.$$

Also, $a^{-1} \neq e$ whenever $a \neq e$. Therefore the $2n - 1$ non-identity elements can be grouped into disjoint pairs

$$\{a, a^{-1}\},$$

each containing exactly two distinct elements.

Consequently, $2n - 1$ would have to be even. But $2n - 1$ is odd, a contradiction. Hence there exists some $a \neq e$ such that

$$a = a^{-1}.$$

Multiplying by a gives

$$a^2 = aa^{-1} = e$$

by the inverse axiom. Thus there exists $a \neq e$ satisfying $a^2 = e$.

Subgroups

Definition 2.2 (Subgroup). Let $(G, *)$ be a group. A non-empty subset H of G is called a **subgroup** of G if H itself forms a group under the same operation $*$ defined on G .

Example 2.2. Consider the group $(\mathbb{Z}, +)$ and let

$$H = 2\mathbb{Z} = \{\dots, -6, -4, -2, 0, 2, 4, 6, \dots\},$$

then show that H is a subgroup of $(\mathbb{Z}, +)$.

Solution

We show that H is a subgroup of $(\mathbb{Z}, +)$.

1. Closure:

Let $a, b \in H$. Then $a = 2m, b = 2n$ for some $m, n \in \mathbb{Z}$. Hence $a + b = 2m + 2n = 2(m + n)$. Since $m + n \in \mathbb{Z}$, we have $a + b \in H$. Therefore, H is closed under addition.

2. Associativity:

For $a, b, c \in H$, $(a + b) + c = a + (b + c)$, because addition in $\mathbb{Z}$ is associative.

3. Identity:

Now, for every $a \in H$, $a + 0 = 0 + a = a$. Hence, 0 is the identity element in H .

4. Inverse:

Let $a \in H$. Then $a = 2m$ for some $m \in \mathbb{Z}$. The additive inverse of a is $-a = -2m \in H$.

Hence $H = 2\mathbb{Z}$ is a subgroup of $(\mathbb{Z}, +)$.

Example 2.3. Consider the group $(\mathbb{R} \setminus \{0\}, \cdot)$ under multiplication. Let $H = \{1, -1\}$, then H is a subgroup of $(\mathbb{R} \setminus \{0\}, \cdot)$.

Example 2.4 (Non-example). Consider the group $(\mathbb{Z}, +)$ and the subset

$$H = \{0, 1, 2, 3, \dots\}.$$

Although H is closed under addition and contains the identity element 0, it does not contain the additive inverse of every element.
Hence H is **not** a subgroup of $(\mathbb{Z}, +)$.